

# SOFTWARE AS A SERVICE

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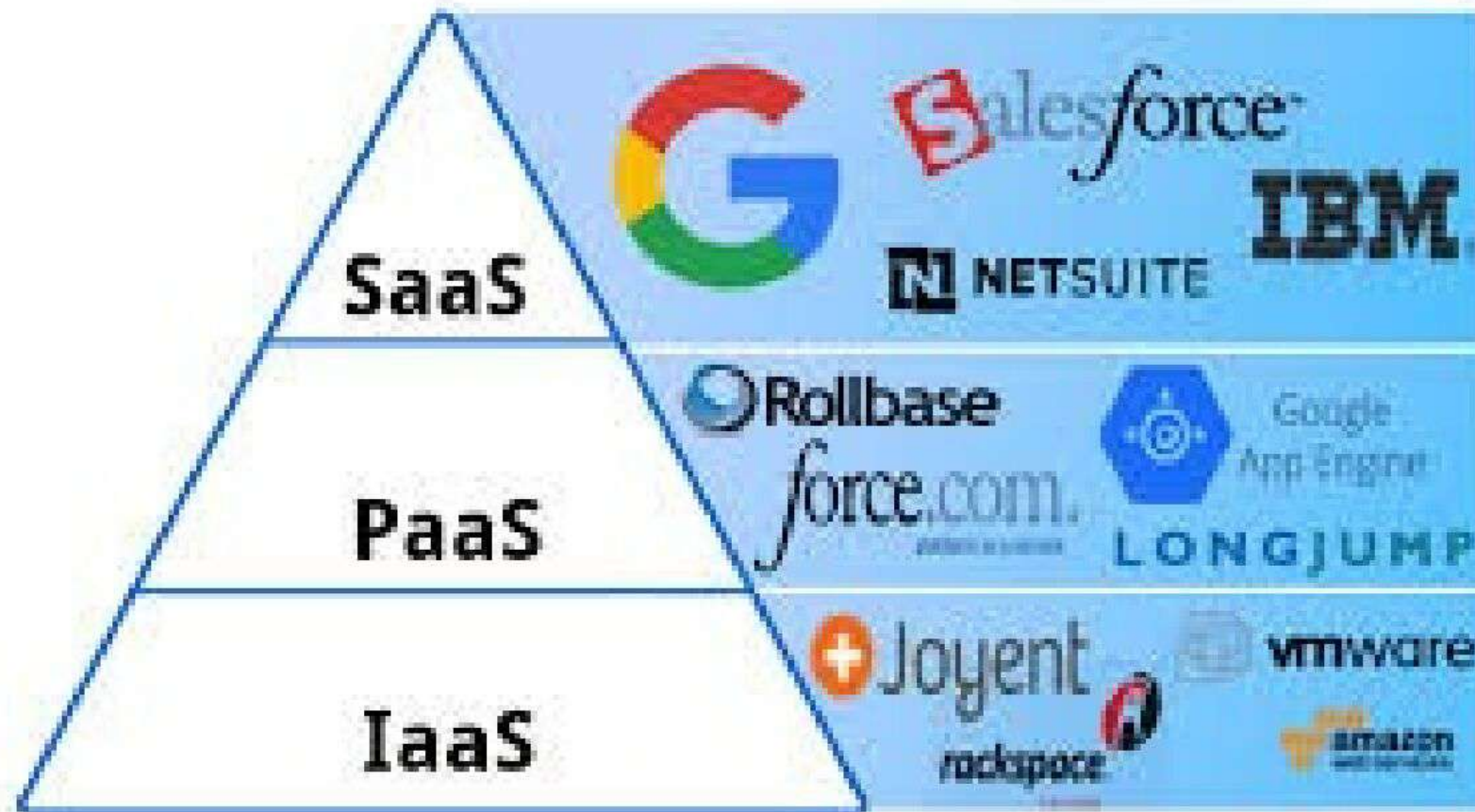
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# SOFTWARE AS A SERVICE



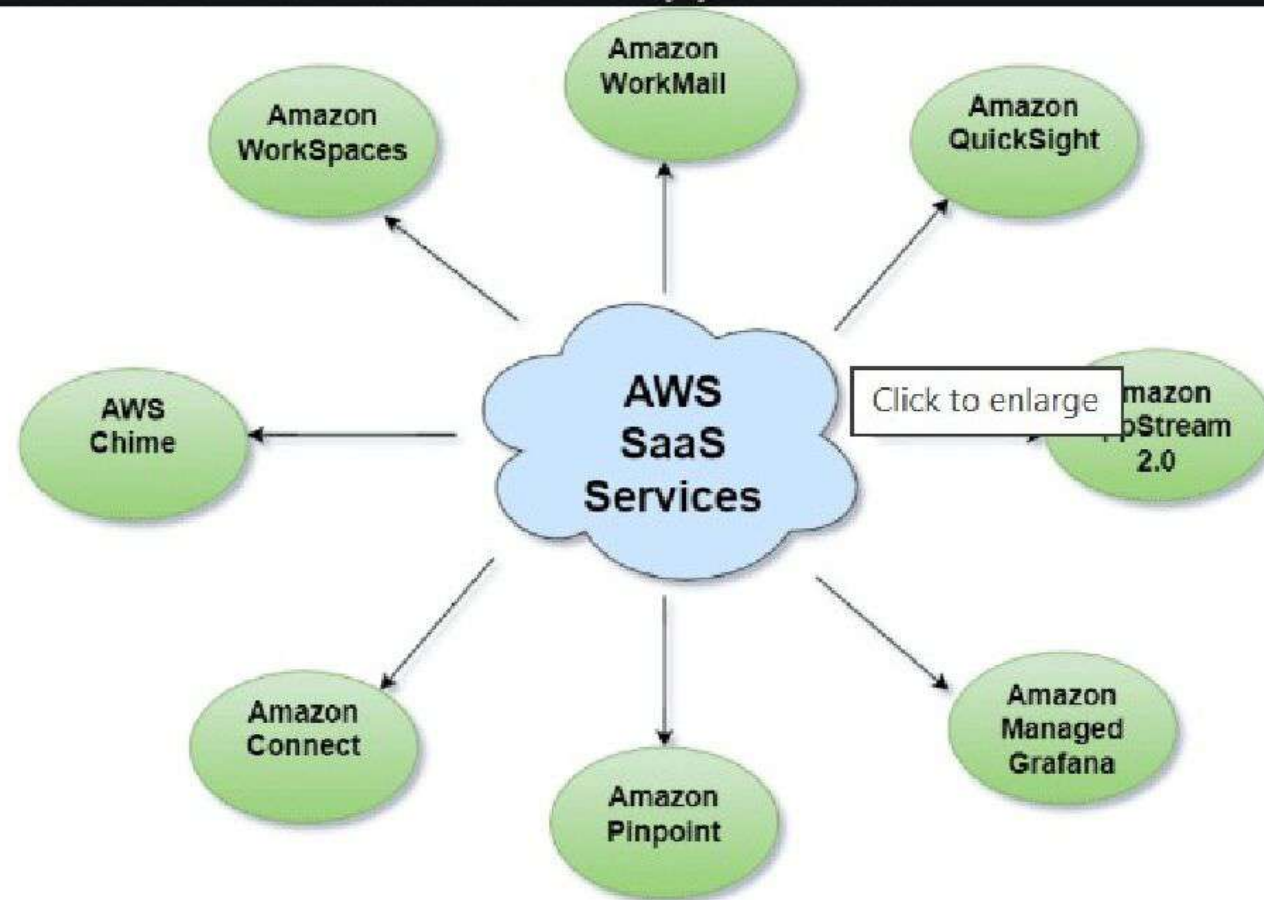
# What is SaaS?

Software as a Service (SaaS) is a cloud-based model where users subscribe to and access applications over the internet via web browsers, rather than installing them locally





## Top AWS SaaS Solutions for Modern Cloud Applications



AWS SaaS Services

## Traditional Software vs SaaS

### Traditional Software

Installed on local PC

High upfront cost

Manual updates

Device dependent

Limited scalability

### SaaS

Runs on cloud

Subscription-based

Automatic updates

Accessible anywhere

Highly scalable

## Software as a Service | SaaS

SaaS is also known as "**On-Demand Software**". It is a software distribution model in which services are hosted by a cloud service provider. These services are available to end-users over the internet so, the end-users do not need to install any software on their devices to access these services.

# SaaS Services





## Popular SaaS Providers



## Key Characteristics of SaaS

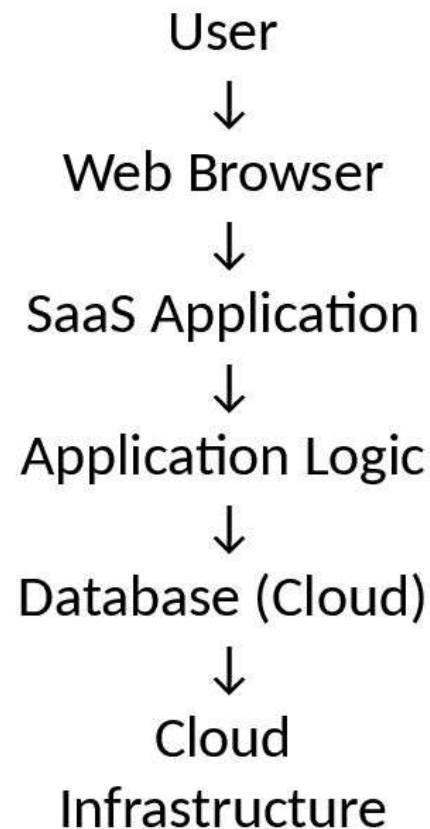
- ❖ Web browser based access
- ❖ Multi-tenant architecture
- ❖ Centralized data storage
- ❖ Automatic updates
- ❖ High availability
- ❖ Subscription pricing

## Why SaaS is Popular?

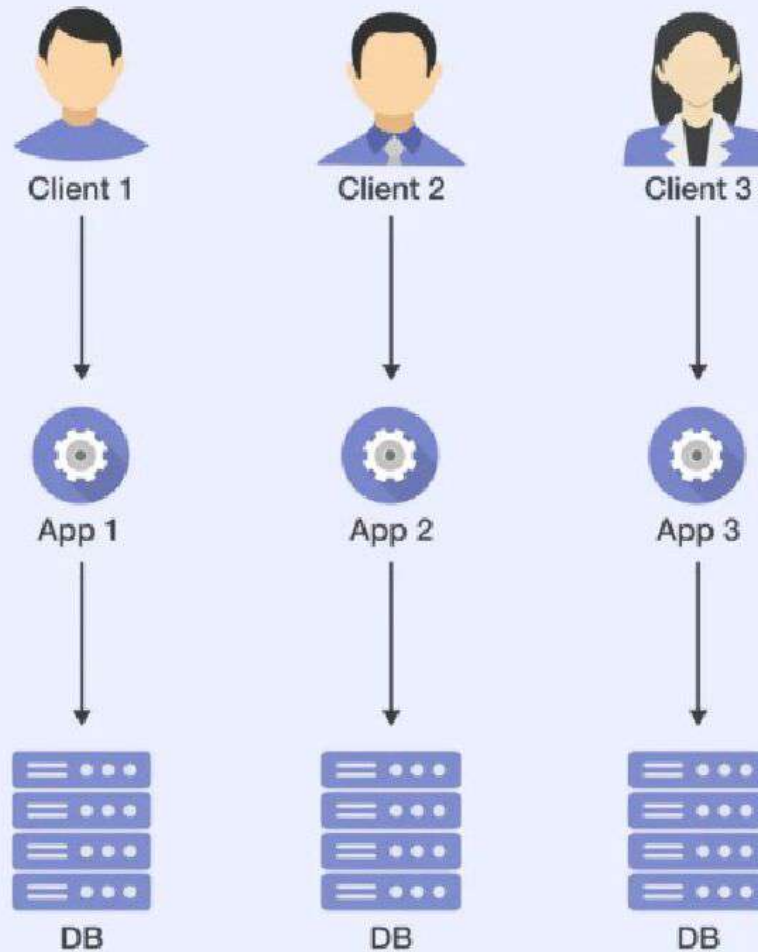
- ❖ No installation
- ❖ No maintenance
- ❖ Lower cost
- ❖ Easy collaboration
- ❖ Anywhere access

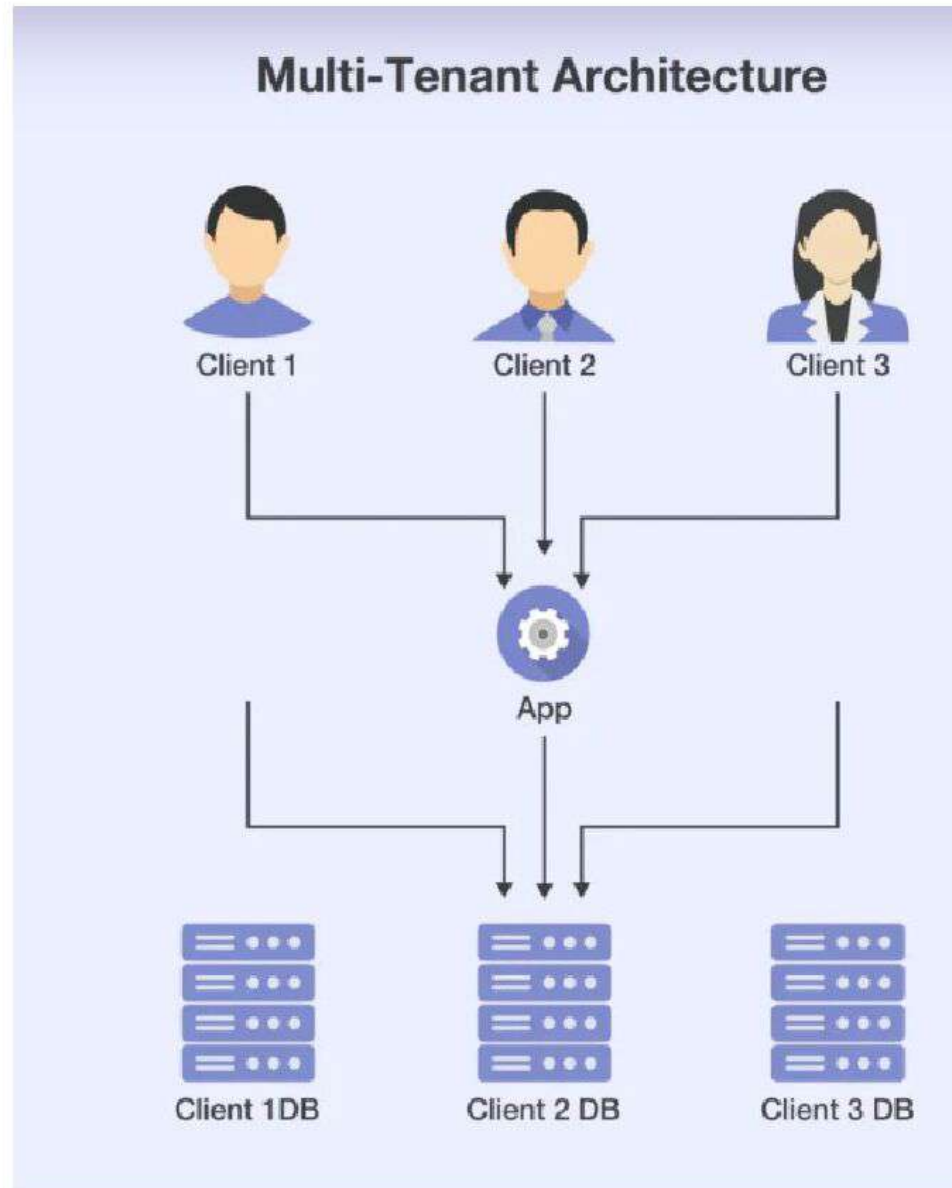
# SaaS ARCHITECTURE & CHARACTERISTICS

## SaaS Architecture



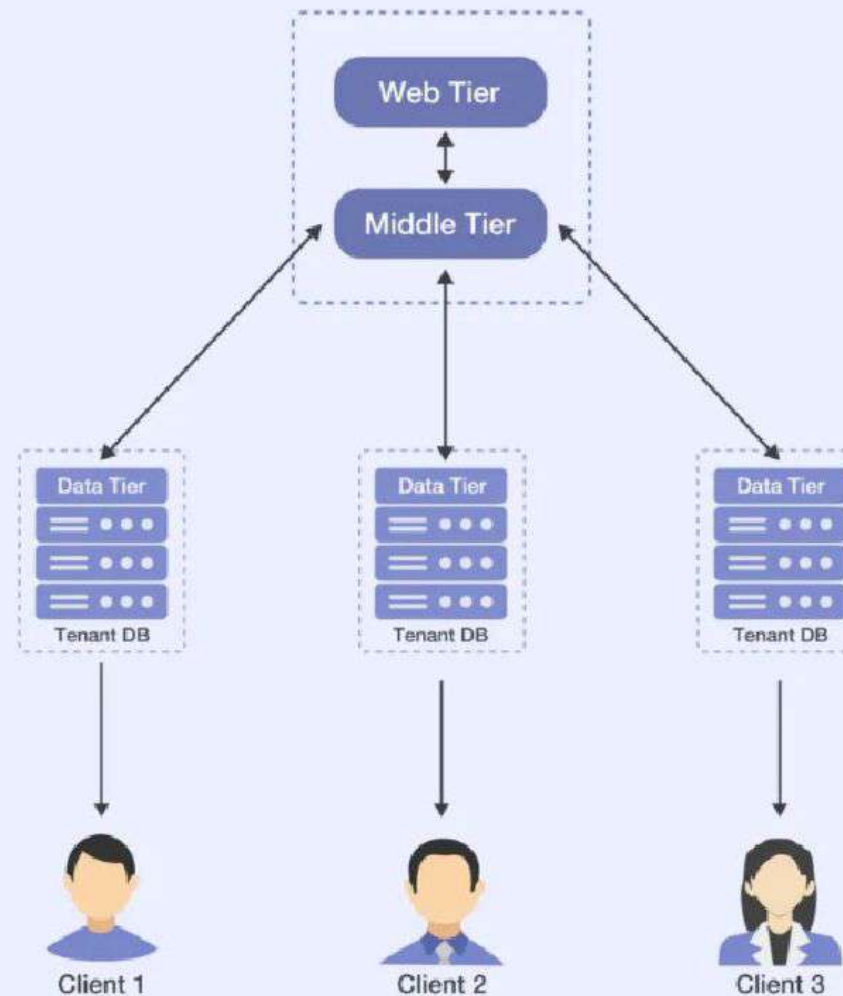
### Single-Tenant Architecture







### Mixed-Tenant Architecture



# SaaS Components

User Interface (Browser / App)

Security & Authentication

Application Logic

Middleware

Database

Cloud Infrastructure

## Stages of the SaaS Software Development Life Cycle



# **WEB SERVICES**

## **(FOUNDATION OF SaaS)**

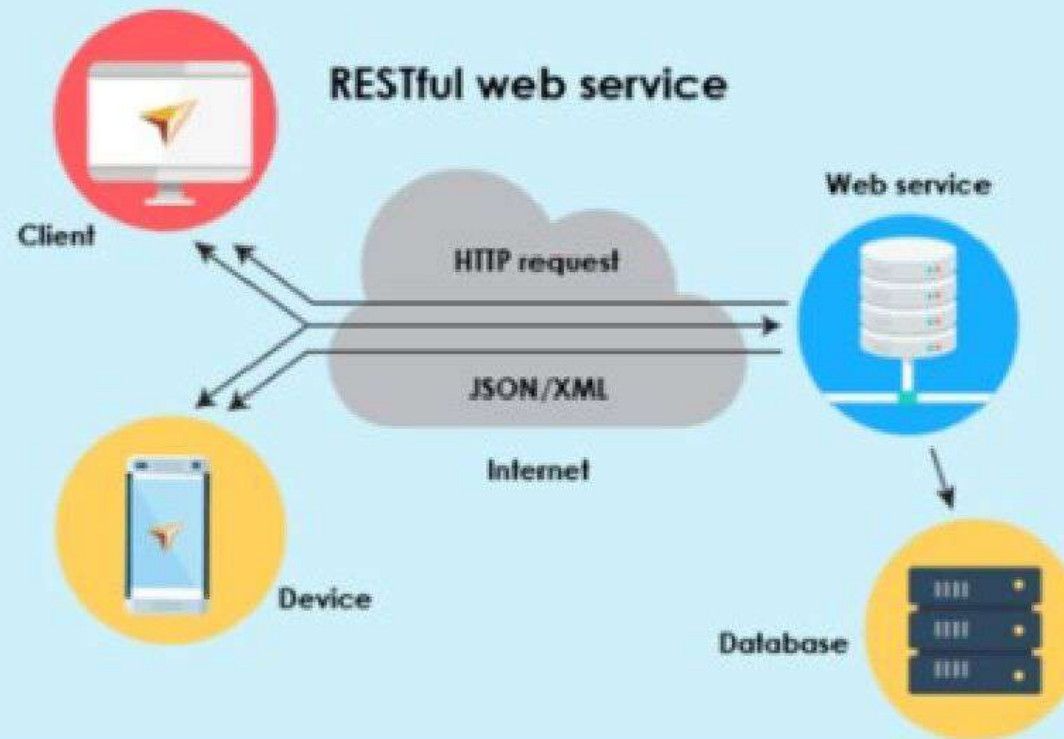
What are Web Services?

Web services are software systems designed to support communication between applications over the web.

SaaS uses web services to:

- Exchange data
- Integrate services
- Enable APIs

# Web Service & API Testing





## **Types of Web Services**

### **1. SOAP (Simple Object Access Protocol)**

XML-based

Heavy but secure

Used in enterprise apps

### **2. REST (Representational State Transfer)**

Lightweight

Uses HTTP methods (GET, POST, PUT, DELETE)

Most common in SaaS

## Role of Web Services in SaaS

Backend communication

Data exchange

Integration with other apps

Mobile app connectivity

Example:

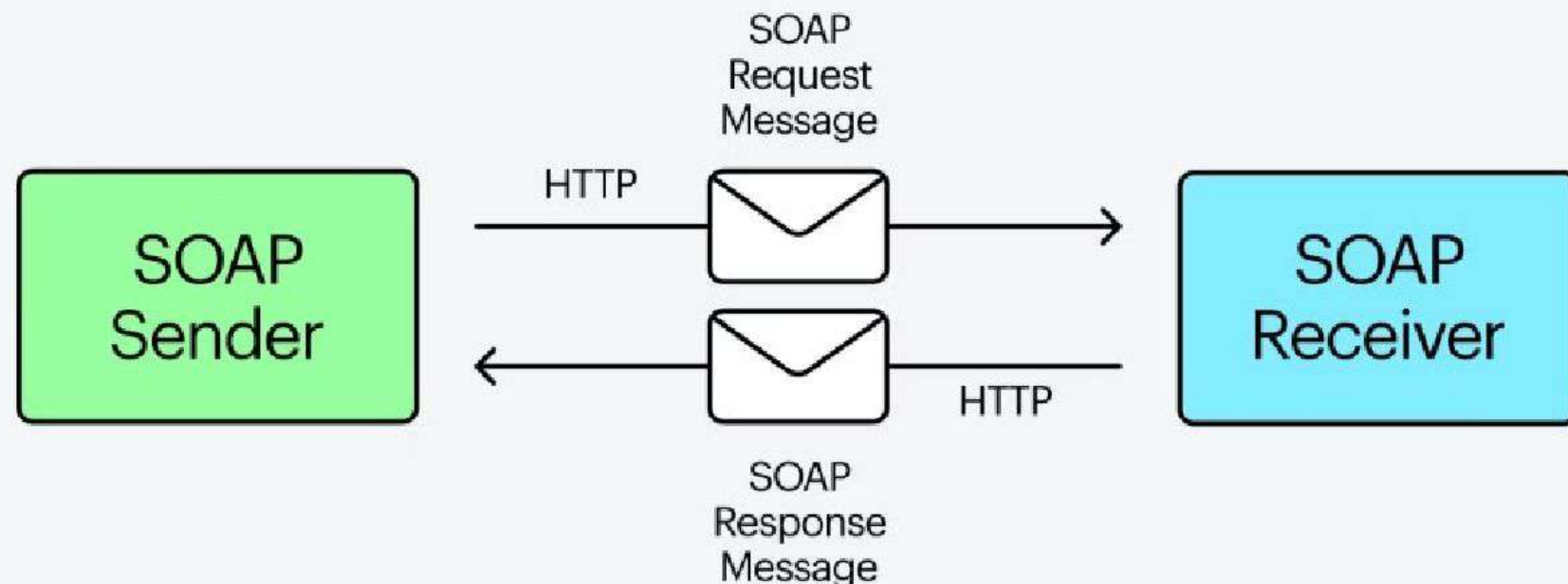
Gmail uses web services to:

Fetch emails

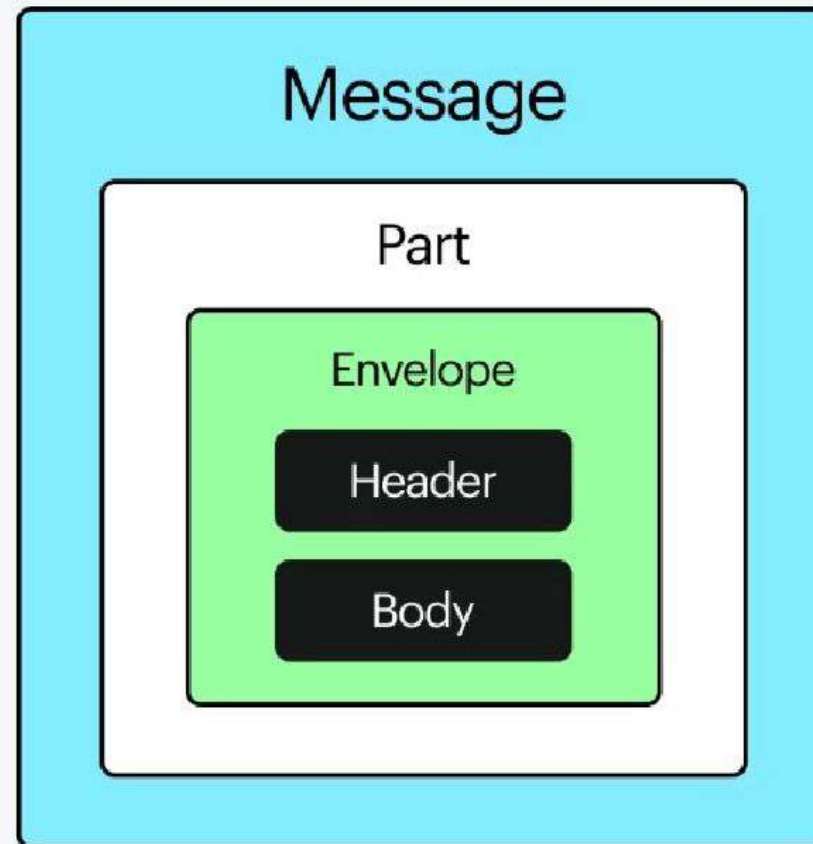
Sync contacts

Send notifications

SOAP (Simple Object Access Protocol) is a messaging protocol used for exchanging structured information between systems over a network, using XML for messaging and HTTP or SMTP for transport.



# Components of SOAP



<https://www.ramotion.com/blog/soap-in-web-services/>

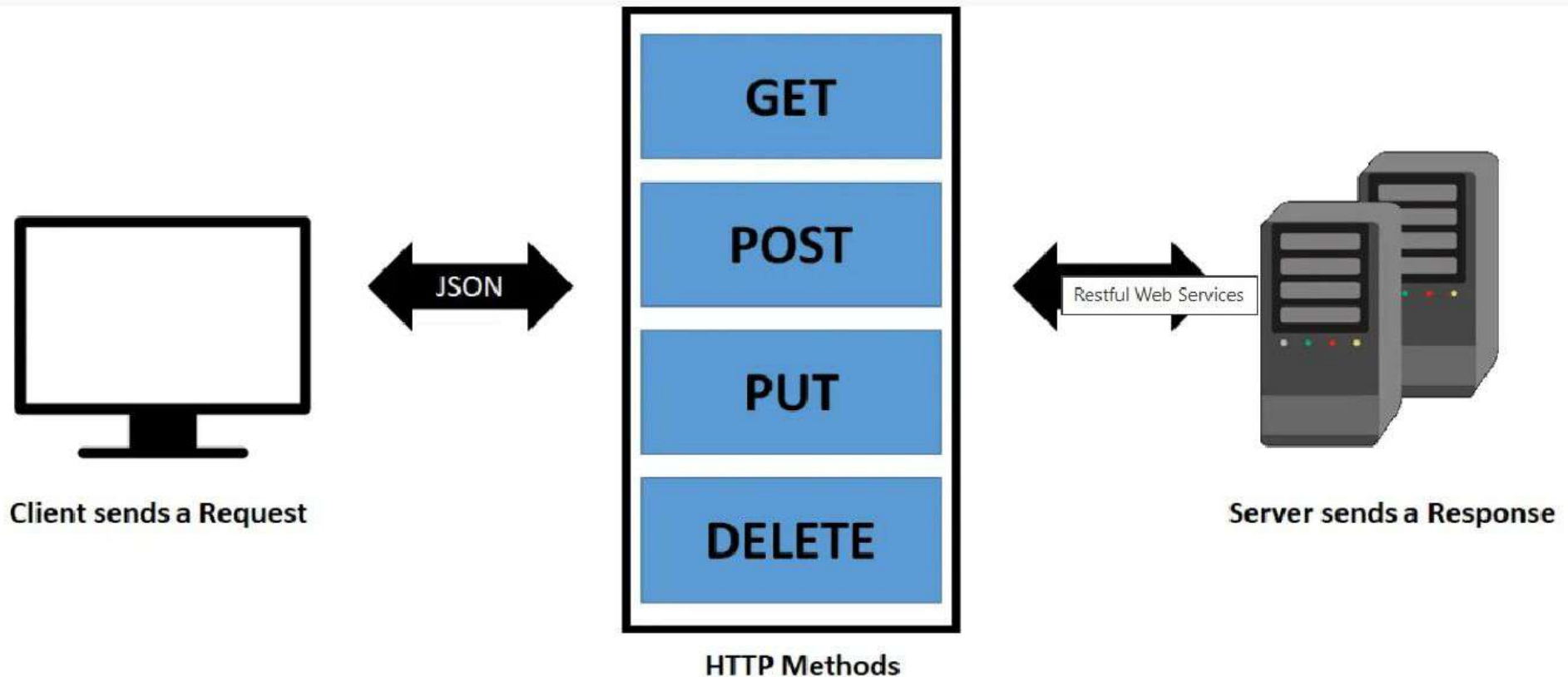
## How to Build SOAP Web Service?

- 1** Determine the requirements of your web service
- 2** Choose a programming language and framework
- 3** Define the WSDL (Web Services Description Language) file
- 4** Implement the web service methods
- 5** Test and Deploy the web service
- 6** Maintain and update the web service



## What is REST

As stated earlier, REST stands for Representational State Transfer. It is a simple way of sending and receiving data between client and server. It doesn't require any software or standards to transfer data. It has a predefined structure to do the API call. Developers just need to use the predefined way and pass their data as JSON payload.



## What is Web 2.0?

Web 2.0 refers to interactive, user-centric web applications that allow collaboration and content sharing.

### Features of Web 2.0

User participation

Rich user interfaces

Social interaction

Dynamic content

APIs & mashups

### Web 2.0 Technologies

AJAX

JavaScript

HTML5

CSS3

APIs

## Relationship Between Web 2.0 and SaaS

### Web 2.0

Interactive UI

User collaboration

APIs

Dynamic content

### SaaS

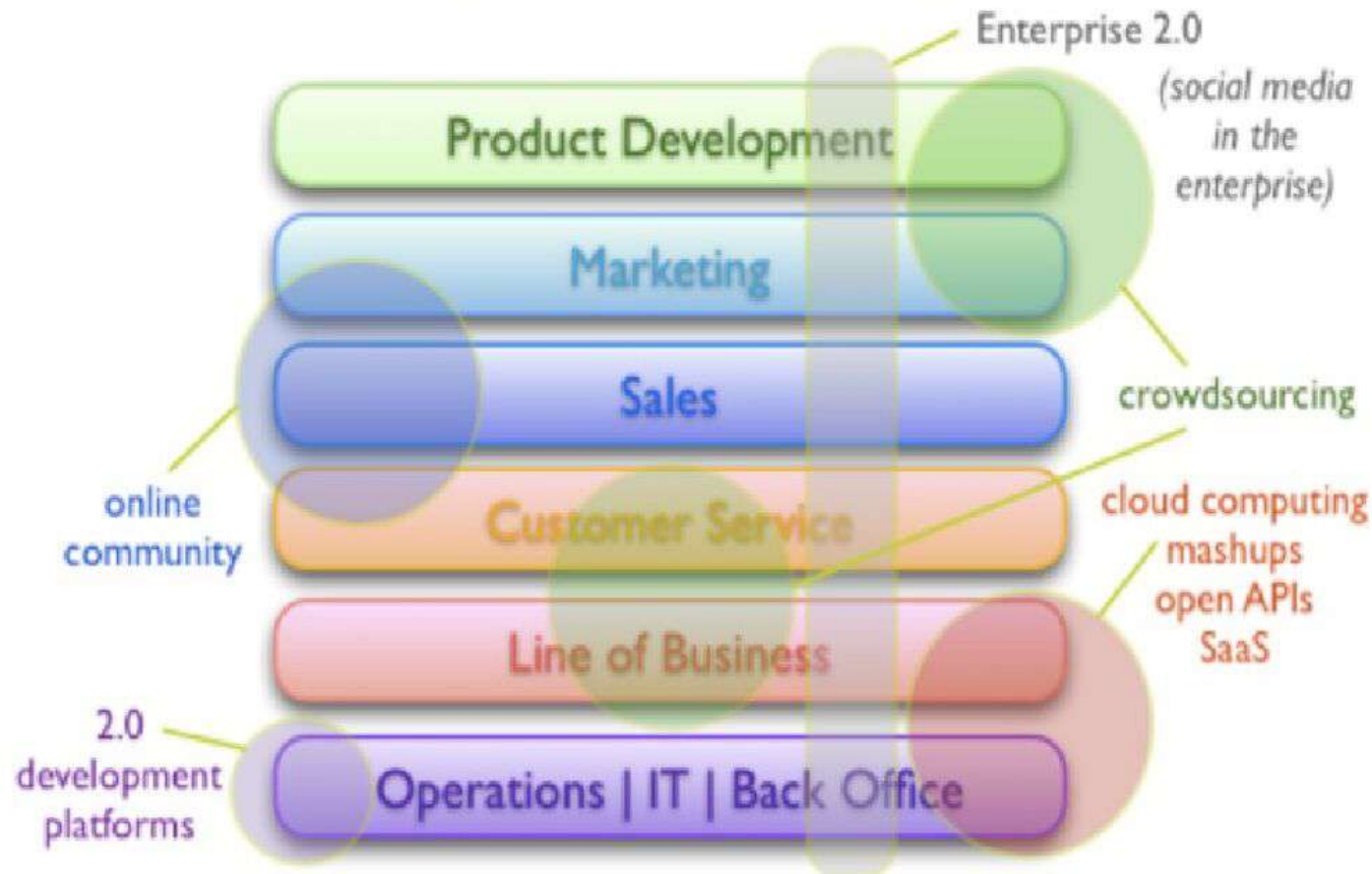
Cloud software

Multi-user access

Web services

Real-time updates

## Strategic Application of Web 2.0 in Business



From <http://blogs.zdnet.com/Hinchcliffe>

## WEB OS & SaaS

A Web OS is an operating system that runs inside a web browser and uses cloud services.

### Web OS vs Traditional OS

| Traditional OS        | Web OS                 |
|-----------------------|------------------------|
| Local installation    | Browser-based          |
| Hardware dependent    | Cloud dependent        |
| Limited collaboration | Built-in collaboration |

### Features of Web OS

Browser-based interface  
Cloud storage  
App-centric  
Device independent

### Role of Web OS in SaaS

Runs SaaS apps smoothly  
Reduces hardware dependency  
Supports remote work

### Examples of Web OS

Chrome OS  
eyeOS  
Cloud-based desktops



# SaaS EXAMPLES & CASE STUDIES

## **Business:**

Salesforce  
Zoho CRM

## **Productivity:**

Google Docs  
Microsoft 365

## **Communication:**

Gmail  
Zoom

## **Education:**

Google Classroom  
Coursera

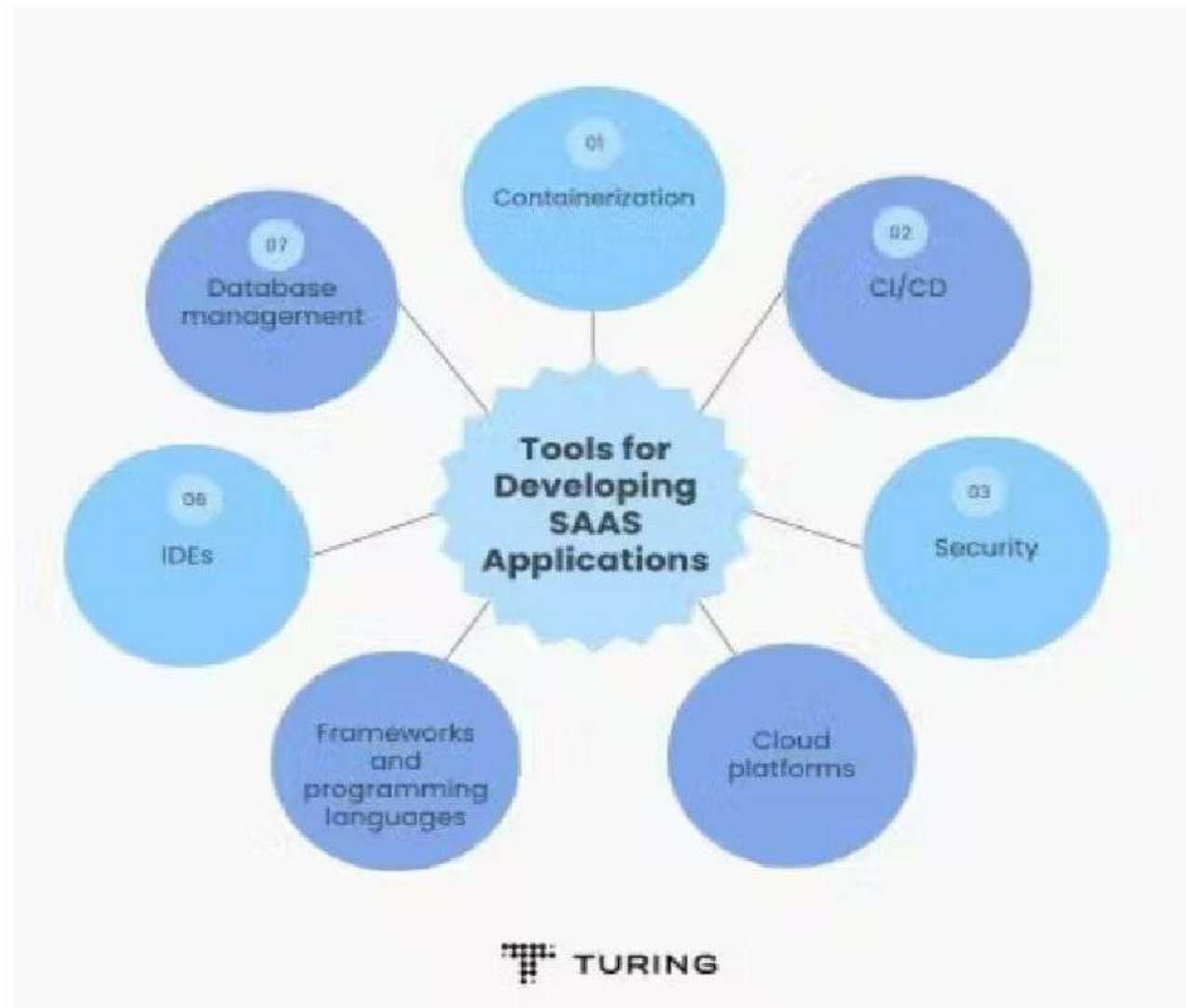
## SaaS in Daily Life

Email  
Online banking  
Video conferencing  
Social networking

## Case Study: Google Docs

No installation  
Real-time collaboration  
Auto-save  
Cloud storage

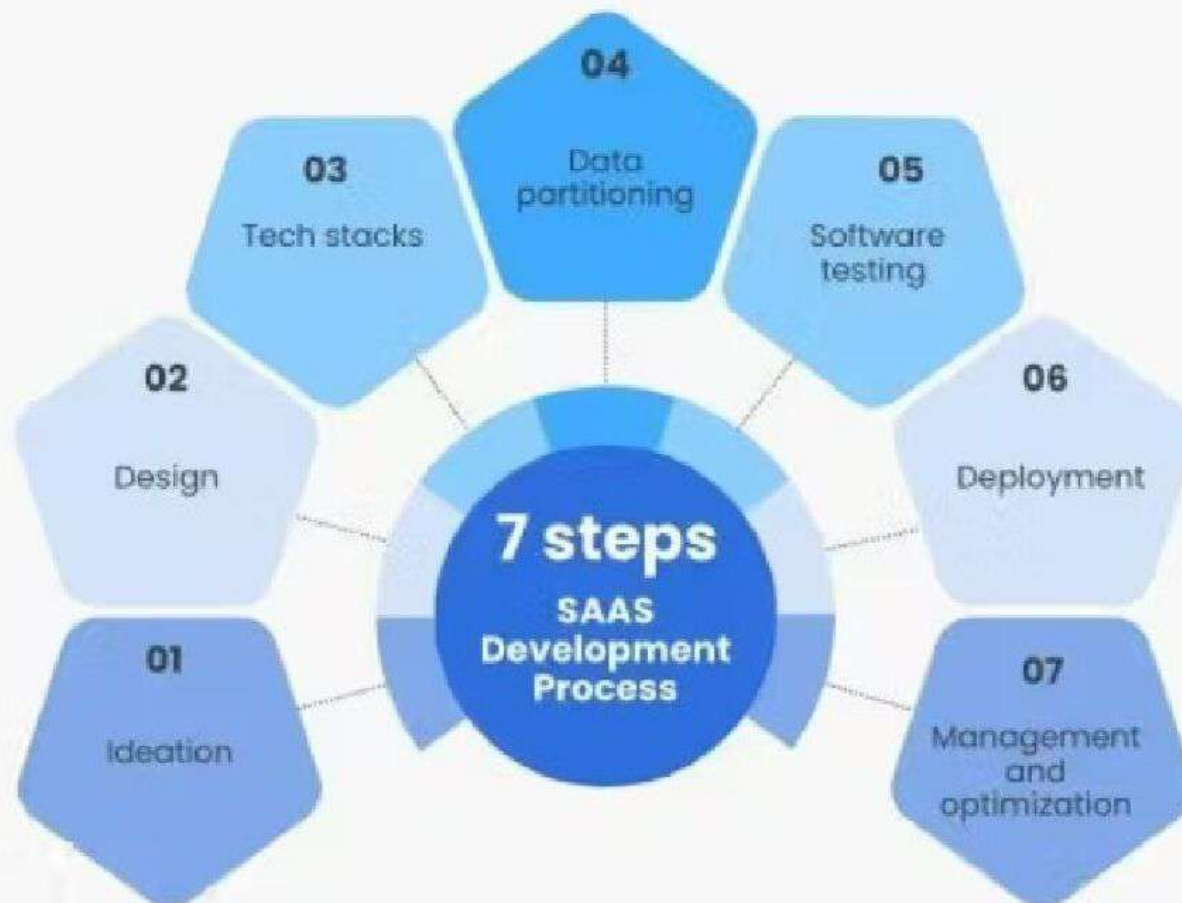
## HOW TO IMPLEMENT SaaS



## Steps to Implement SaaS

1. Identify application
2. Design multi-tenant architecture
3. Develop web-based UI
4. Use web services/APIs
5. Deploy on cloud platform
6. Implement authentication
7. Enable billing & monitoring

# SaaS development process



TURING

User → Browser → SaaS App → Web Services → Cloud DB



# SECURITY, ADVANTAGES, LIMITATIONS

User **never** installs software

Access via **URL**

Updates handled by provider

## Security in SaaS

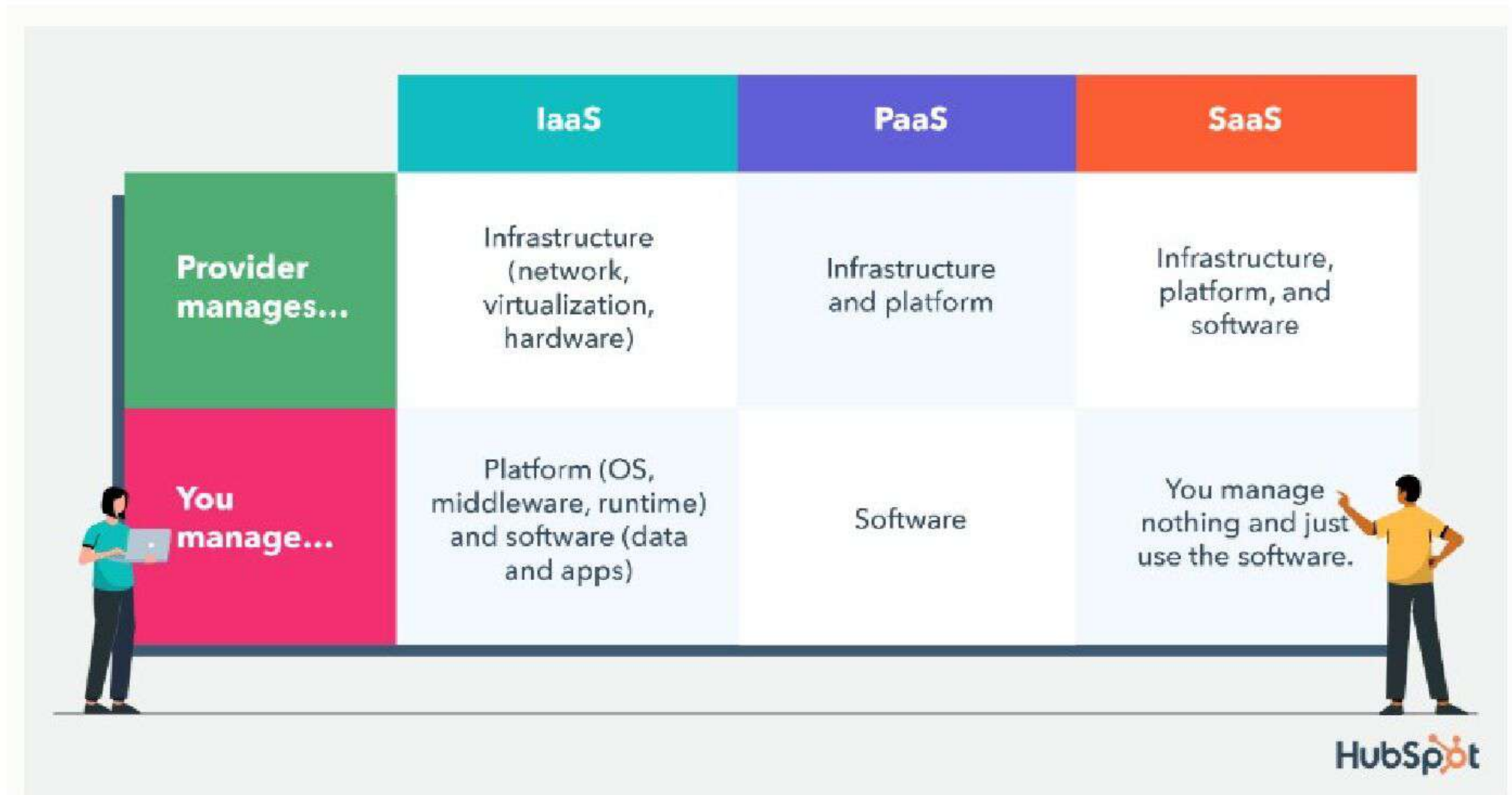
- Authentication
- Authorization
- Data encryption
- Secure APIs
- Regular backups

## Advantages of SaaS

- ✓ Cost effective
- ✓ Easy access
- ✓ Automatic updates
- ✓ High availability
- ✓ Scalability

## Limitations of SaaS

- Internet dependency
- Limited customization
- Data security concerns
- Vendor lock-in



## Characteristics of IaaS, PaaS, and SaaS

### IaaS

Scalability with on-demand resources

Cost-effective: Only pay for the resources used

High level of customization offered

Robust security measures such as firewalls, etc.

### PaaS

Suite of developer tools and services offered

Middleware services provided

Automatically scales resources

Easy collaboration on projects

### SaaS

Accessible from any device

Automatic updates for features and security

Subscription-based pricing

Integration with other cloud services



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